YULIIA STANENKO

AI & FULL STACK SOFTWARE ENGINEER

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PROFILE

Software engineer with 2+ years of experience building production web and mobile applications with TypeScript, React, Next.js, React Native, Node.js and PostgreSQL, now focused on applied AI with Python. Builds LLM-powered systems end to end: Retrieval-Augmented Generation (RAG), structured LLM outputs, classification models, and evaluation that proves quality - labelled evaluation sets, LLM-as-judge, automated evals in CI and drift monitoring. Comfortable across the stack: system design, REST APIs, integrations, relational data modelling, authentication and security, with two apps shipped to TestFlight for a Swedish healthtech company. Currently an AI/ML Research Intern at Lund University. Works closely with users and domain experts, turning ambiguous problems into practical, well-tested software.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL
AI/ML: Large Language Models, Generative AI, RAG (hybrid retrieval, embeddings, reranking), prompt and context engineering, structured JSON outputs, classification models, local LLMs (Ollama, llama.cpp), vision-language models (Gemma, Llama), Hugging Face models (sentence-transformers, Whisper, pyannote), scikit-learn, LightGBM
Evaluation & LLMOps: labelled evaluation sets, LLM-as-judge, F1 and Cohen's kappa, prompt versioning via pull requests, automated evaluation in CI, drift monitoring, model comparison (quality, latency, cost)
Frontend: React, Next.js, React Native (Expo), Zustand, Tailwind CSS, Recharts, i18next
Backend: Node.js / Express, Hono, REST API design, system integrations, background workers, Socket.IO, Zod validation
Databases: PostgreSQL (Neon, pgvector), SQL, Prisma ORM, MySQL/MariaDB, MongoDB, Redis, Firebase Firestore
Security: JWT, bcrypt, two-factor auth (TOTP), rate limiting, Helmet, protected routes, privacy-first data processing
Cloud, CI/CD & Tools: AWS S3, Cloudflare (Workers AI, R2), Vercel, Railway, Firebase, Vitest, GitHub Actions, Git / GitHub, Figma
Spoken languages: English: advanced, Ukrainian: bilingual, Russian: bilingual, Swedish: basic

EDUCATION

Master of Science in specialty "Computer Science" **2024-2025**
National Technical University "Kharkiv Polytechnic Institute", Ukraine
Specialization: Mathematical Modeling and Intelligent Computing in Engineering

Bachelor's degree in specialty "Software Development" **2020-2024**
Zaporizhzhya National University, Ukraine

WORK EXPERIENCE

AI/ML Research Intern, Lund University (Sweden) **11.2026-present**

- Working on a research project on the analysis of conversational behaviour: multi-label classification of behaviour patterns in conversation transcripts (e.g. evasive, challenging or non-responsive answers), plus impact and engagement ratings, with supporting quotes and confidence for each judgement.
- Operationalising how human annotators make decisions: refining the annotation protocol, labelling first examples and translating annotator judgements into measurable text and conversation features.
- Designing a privacy-first Python pipeline that runs fully locally, with no cloud APIs: transcripts with speaker diarisation (Whisper, pyannote), trained feature-based classification models (scikit-learn, LightGBM, sentence-transformers) and open-source LLMs (Ollama / llama.cpp) returning structured JSON, from zero-shot to few-shot prompting.
- Planning evaluation against human annotators (F1, Cohen’s kappa, weighted kappa), including LLM output stability and checks that cited quotes actually appear in the transcript.

Full Stack Developer, GlobePerson & GlobePatient, HEdVC (Sweden) **03.2025-03.2026**

- Independently designed and built all frontend screens of a cross-platform mobile app and connected them to two parallel backend services.
- Implemented core product features end to end - authentication, notifications, multi-language support, and file uploads.
- Debugged performance and stability issues across devices, testing on physical hardware throughout development.

Accelerator Program Winner and App Development (Ukraine, Walkey app) **10.2024-07.2025**

- Won an accelerator by building Walkey end to end, from concept to a working, deployed product, under tight time constraints.
- Owned system architecture and the matching-algorithm design; chose and integrated the stack (Expo Router, Clerk, Neon, Firebase, Socket.IO); implemented core features - matching, real-time location, chat, and notifications.
- Collaborated on product concept and UX with the team, contributing to the winning solution.

FEATURED PROJECTS

DocTrace - RAG assistant for regulatory documents with verified citations (in progress)
Python, PyMuPDF, fastembed (multilingual-e5-large), Postgres + pgvector (Neon), FastAPI, LangGraph, Next.js, Docker, GitHub Actions

- Building a Retrieval-Augmented Generation (RAG) assistant that answers questions over long documents (EU AI Act, GDPR) with citations to exact clauses and never invents sources; the corpus is swappable.
- Built a structure-aware ingestion pipeline that turns PDFs into 981 structured chunks with exact references (e.g. AI Act, Article 5(1), page 51) and self-checks against expected structure (113/113 articles, 180/180 recitals).
- Designed hybrid retrieval in a single Postgres database: pgvector embeddings (multilingual, so a question in Russian finds English text) plus full-text search for exact terms, with cross-encoder reranking to the top 5.
- Grounded answers by design: the LLM returns structured JSON, deterministic code verifies every quote verbatim against its source chunk, and the system answers "not found" when the documents do not contain the answer.
- Next: LangGraph agent with search, article comparison and requirement extraction tools, and evaluation on a hand-labelled question set (Recall@5, MRR, citation precision) running in GitHub Actions.

StyleGuard - brand consistency control for AI-generated images
TypeScript, Hono, Neon Postgres, Cloudflare R2, Cloudflare Workers AI, Lovable, GitHub Actions

- Built a Generative AI system that generates images from versioned brand templates (style, rules, palette, pass score) and automatically checks every image for brand consistency; each image is linked to the brand version it was made with.
- Designed an async generation pipeline with a background worker (generate - store in R2 - evaluate - done), with retries, timeouts and duplicate protection.
- Implemented a vision-model judge that scores images 0–10 with an explanation; the verdict is decided by code against a threshold, and invalid model output is never turned into a made-up score.
- Evaluated the judge against 42 manually labelled images: Gemma reached 93% agreement and caught 100% of off-brand images, Llama 79%.
- LLMOps: prompt changes go through pull requests where a GitHub Action runs the evaluation automatically; added drift monitoring with a Slack alert, model comparison by score, latency and cost, and spend limits.

ApplyTrack - Job Application Tracker (personal project)
Next.js 16, TypeScript, PostgreSQL, Prisma, Zod, Tailwind CSS, Vitest, GitHub Actions

- Built a full job-search platform end to end: application tracker with a Kanban board (drag-and-drop, 6 status columns), a timeline of events per application, and follow-up reminders.
- Implemented an auto job-search feed aggregating two external sources (Remotive, Arbeitnow) through a shared adapter interface, with deduplication via composite unique keys.
- Built an email digest system (Resend + Vercel Cron) that matches saved search criteria to new listings and sends daily summaries; logs every send for auditability.
- Cookie-based authentication with bcrypt, httpOnly/secure/sameSite cookies and protected routes; Vitest unit tests and GitHub Actions on every push. Live at applytrack-sooty.vercel.app.

GlobePerson & GlobePatient - Cross-Platform Healthtech App
React Native (Expo), TypeScript, Zustand, Node.js/Express, MySQL/MariaDB, AWS S3, Redis, Socket.IO

- Owned the frontend end to end: built every screen and connected the UI to two parallel backend services - an in-house Node.js/Express API and an external Spring Boot (Java) service on Tomcat.
- Implemented authentication and security (JWT, bcrypt, two-factor auth/TOTP, rate limiting, Helmet), plus push notifications end to end (Expo push tokens, Firebase Admin, deep-link navigation).
- Built image/document upload flows with files stored in AWS S3 and a MySQL/MariaDB database on AWS; real-time updates via Socket.IO with Redis caching; multi-language support with i18next.

Walkey - Dog-Matching & Social Walking App
React Native (Expo Router), TypeScript, Zustand, NativeWind, Clerk, Node.js/Express, Neon (Postgres), Firebase, Socket.IO, AWS

- Designed and implemented a custom weighted matching algorithm scoring compatibility across eight attributes, with geographic filtering via the Haversine formula.
- Implemented real-time location tracking (expo-location + Socket.IO) with a privacy layer that shares only proximity - never a user's exact home address.
- Built real-time chat, walk scheduling and push notifications (Firebase Cloud Messaging); authentication with Clerk alongside a custom Express backend.

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